

SQL DAY 3 – Main Queries Explanation

This SQL file starts with table creation and data insertion queries. These tables and inserted values are specifically created to demonstrate the use of **JOIN operations** and **FOREIGN KEY relationships** between tables.

1. CREATE TABLE Query

The CREATE TABLE query is used to define the structure of the table. In this file, tables are created with primary keys so they can be referenced later using foreign keys.

2. INSERT INTO Query

The INSERT INTO query is used to add data into the tables. The inserted values are meaningful and are used later to perform JOIN operations between related tables.

3. SELECT Query

The SELECT query is used to retrieve data from the table. Using '*' fetches all columns from the table.

JOIN Queries (Brief Explanation)

LEFT JOIN: Returns all records from the left table and matching records from the right table. If no match exists, NULL values are returned.

RIGHT JOIN: Returns all records from the right table and matching records from the left table. If no match exists, NULL values are returned.

Important SQL Topics (Short Explanation)

AUTO COMMIT: Automatically saves every SQL statement. No manual COMMIT is required unless auto-commit is disabled.

PRIMARY KEY: A column that uniquely identifies each record in a table. It does not allow NULL or duplicate values.

FOREIGN KEY: A column that creates a relationship between two tables by referencing the primary key of another table.

AUTO INCREMENT: Automatically generates a new numeric value for each inserted record. Commonly used with primary keys.

UNIQUE: Ensures that all values in a column are different. Duplicate values are not allowed.